

Technical Report

BIG DATA ANALYTICS FINAL PROJECT: Distributed Multi-Model Analytics for E-commerce Data

1. Introduction

This project analyzes an e-commerce platform to identify customer behavior and performance trends, comparing three databases for processing efficiency, document flexibility, time-based lookups, and distributed data processing.

The dataset spans 90 days and includes 10,000 users, 5,000 products, 25 categories, 500,000 transactions, and 2 million sessions.

2. System Architecture

After generating 24 JSON files (users, products, categories, transactions) and 20 session files, the data was loaded into MongoDB and HBase. Key queries included top products by revenue, user segmentation by purchase frequency, and per-user session analysis. Spark was used for distributed workflows, including cohort analysis of user purchases over time. Results were visualized with matplotlib and seaborn.

MongoDB enabled flexible, embedded storage for transaction data. HBase's column-based structure and row-key identifiers supported efficient session and product lookups. Spark processed large data volumes, joining tables to reveal user and product patterns. Python visualization libraries were chosen for speed and simplicity.

3. MongoDB Design & Implementation

During the MongoDB design phase, five collections were created after loading the files generated by running `dataset_generator.py`.

- MongoDB collections: Users (user info, geo, registration, last active), Categories (category info, subcategories), Products (details, price history), Sessions (session/user info, device, views, cart, conversion), Transactions (IDs, items, amounts, payment, status).

Embedded documents were necessary due to the multi-valued nature of the stored data and their role within each collection. After creating these collections, three aggregation pipelines were executed:

- Aggregations included: top 10 products by revenue (prod_00154 led), user segmentation by purchase frequency (low: 2,174; medium: 5,252; high: 2,574), and top revenue categories (cat_005 led with \$22M from 40K orders).

4. HBase Design & Implementation

HBase design: user_sessions (user ID + session time, with info, device, activity families) and product_metrics (product ID + date, with info and metrics families).

HBase's composite keys and column families enabled efficient storage and querying for user sessions and product data at scale.

This highlights the comparative justification for using both MongoDB and HBase in this system.

Criteria	MongoDB	HBase
Data Model	Document (Nested flexible)	Wide - column (sparse, flat)
Best for	Transactions, products, categories	Time - series session data
Query style	Aggregation pipelines	Row key scans, prefix scans
Schema flexibility	High flexibility allowing several variable fields per document	Moderate because of fixed column families
Scalability	Horizontal via sharding	Horizontal via region servers

100,000 session records were loaded into user_sessions and 350,000 purchases were loaded into product_metrics for analysis, enabling queries such as:

- Sessions per user (e.g., user_000042: 11 sessions)
- Daily product performance (e.g., prod_00154: 85 days, 578 units, \$343K revenue)
- Units sold, daily revenue, and order counts per product.

5. Apache Spark Processing

Spark processing was limited by hardware (8GB RAM), so session data was sampled. Other files were loaded and processed successfully.

Nearly half of transactions lacked sessions due to two data generation paths: one from sessions, one randomly to reach 500,000 transactions.

- Spark analyses included: Product affinity: The most frequently co-purchased products were prod_02483 and prod_02898.
- Cohort analysis: Users grouped by registration month; 7 cohorts identified, avg. spend \$920–\$948.
- Payment analysis: PayPal was the most used; bank transfer had the highest average revenue per transaction.
- Revenue by country: LA and CZ led.
- Compared product pairs viewed vs. purchased together.

To address this question, products viewed together in the same browsing session and those purchased together in the same order were analyzed and compared.

Product pairs within sessions and transactions were identified, counted, and compared to find pairs both viewed and purchased together.

Analysis (100,000 sessions, 500,000 transactions, 570,929 pairs): Product pairs viewed together once had a 7.5% chance of also being purchased together, rising to 12.5% when viewed together three times, indicating co-viewing predicts purchase intent.

6. Integration Analysis

This section investigates whether products frequently viewed together by users are also purchased together, or if browsing interest does not translate into sales.

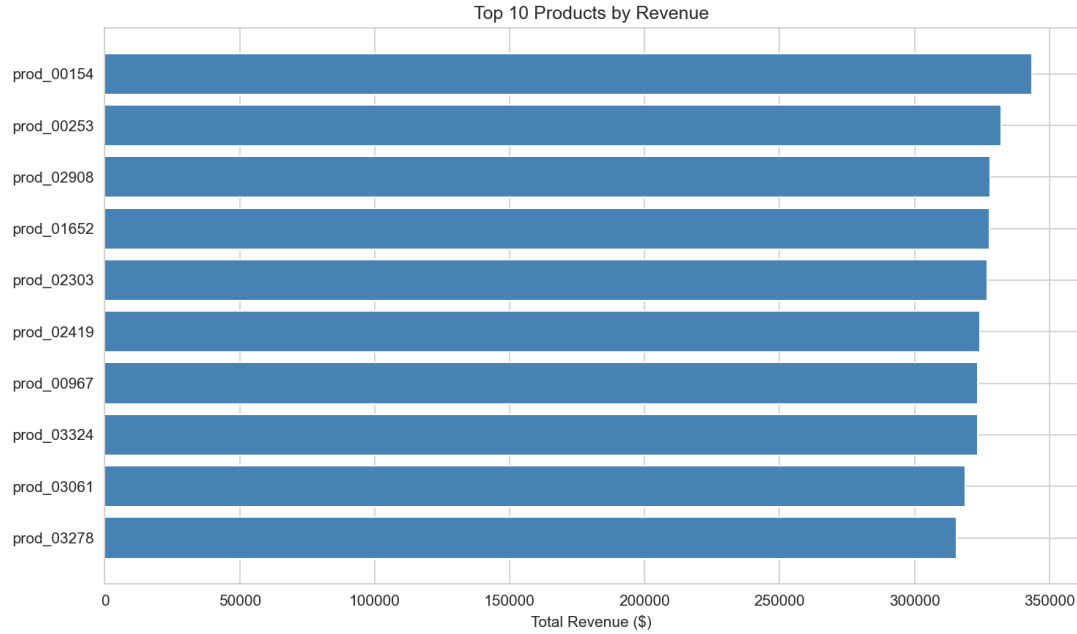
The analysis considered products viewed together within the same browsing session and products purchased together in the same order. Data were processed using Spark to enable comparison. All possible product pairs from sessions or transactions containing more than one product were identified, and the frequency of each pair was calculated.

Subsequently, the data on items viewed together and items bought together were combined to identify product pairs that were both viewed and subsequently purchased together.

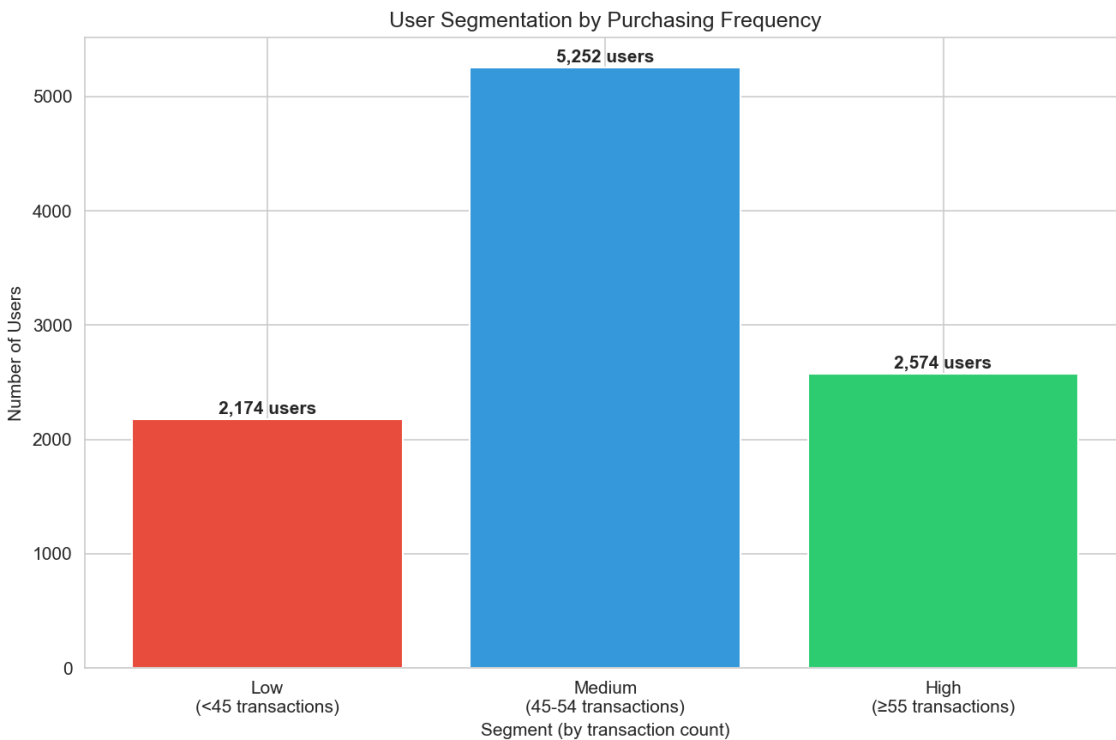
The analysis used a sample of 100,000 sessions and all 500,000 transactions, yielding 570,929 product pairs. The findings indicate that products viewed together are frequently purchased together. Product pairs viewed together only once were purchased together approximately 7.5% of the time. When pairs were viewed together three times, the purchase rate increased to approximately 12.5%. These results suggest that repeated co-viewing is a strong indicator of genuine purchase intent rather than random browsing.

7. Visualizations

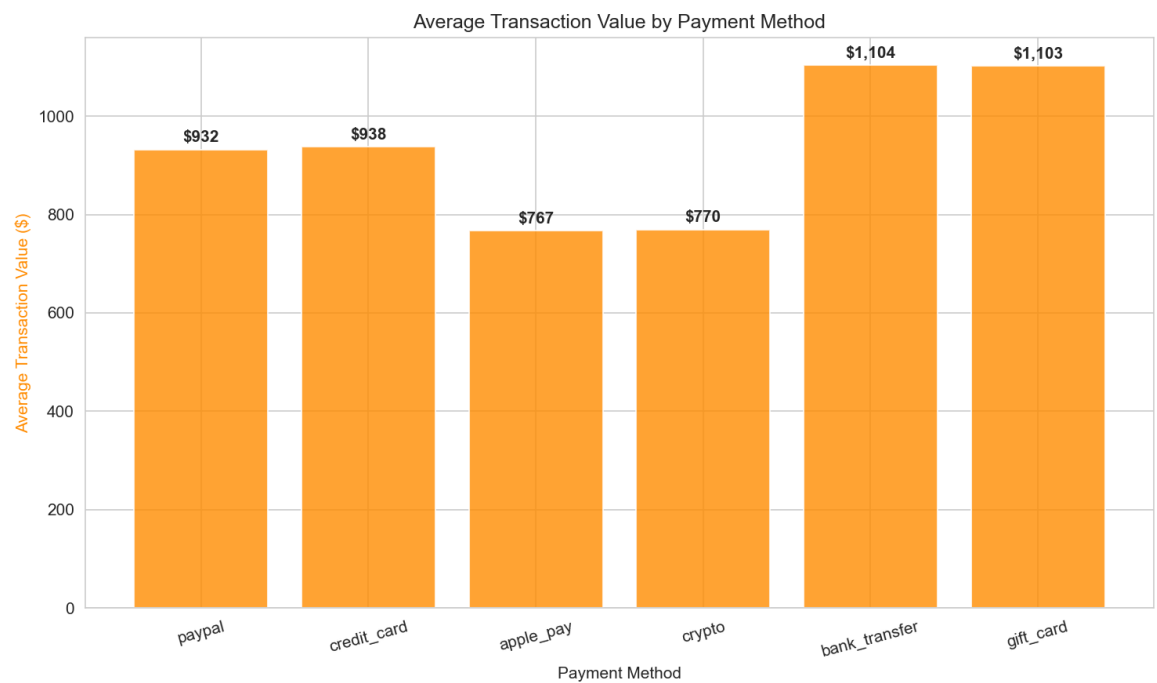
This visualization illustrates the revenue generated by each product.



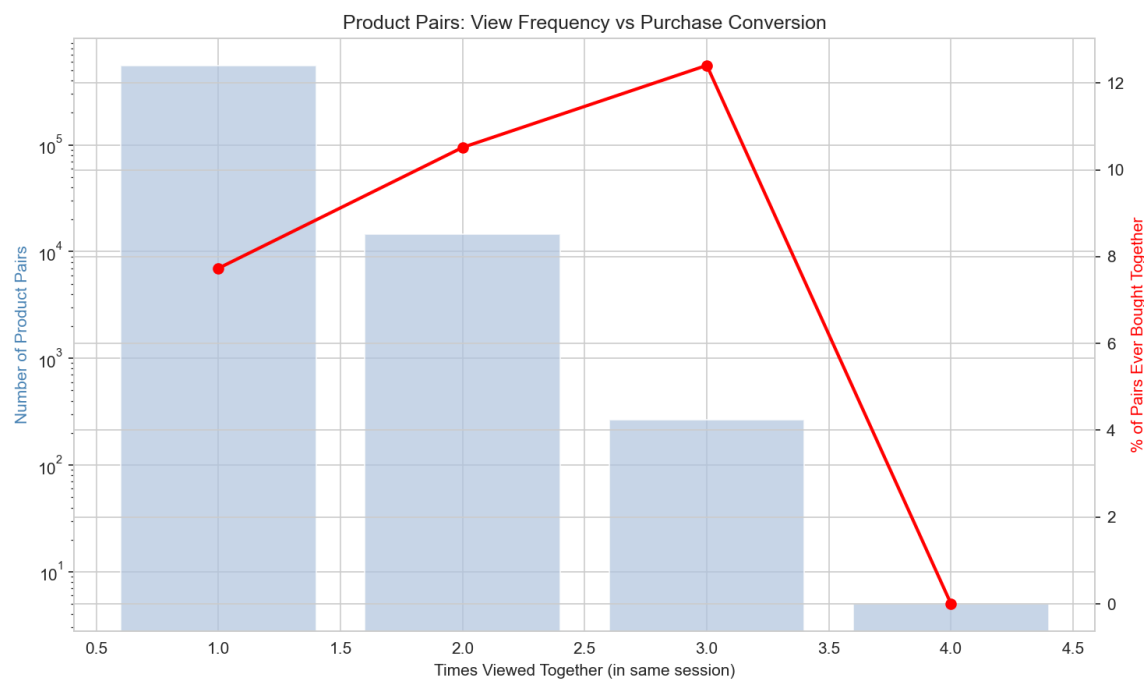
Visualization 2 presents user segmentation by purchase frequency and transaction value by payment method.



Visualization 3 shows the average transaction value according to their payment method.



Visualization 4 compares product pairs based on their view and purchase frequencies.



8. Limitations & Future Work

This project faced several key limitations. Limited hardware, specifically 8GB of RAM, meant that analysis had to be conducted on smaller data samples rather than the full dataset.

Additionally, using synthetic data rather than real user data yielded patterns that were less realistic and more uniform, making the findings more illustrative than practical. There were also duplicate session IDs due to how the dataset was generated, but these were addressed during data processing.

For future work, access to more powerful computing resources would allow for analysis of the complete dataset and eliminate the need for sampling. Incorporating advanced analytics, such as association rule mining, could provide deeper and more actionable insights. Expanding data tracking to cover longer time periods would also enable more comprehensive time-series analysis of product performance.